A Frame-Semantic Deep Learning Approach for Suicide Risk Detection on Social Media

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Abstract

The identification of suicide risk on social media presents a significant challenge for existing models in interpreting metaphorical expressions due to poor semantic understanding. We propose a framework inspired by Cognitive Metaphor Theory that integrates semantic frame features with textual embeddings, using a BiLSTM-Attention network for classification. Experiments show our framework significantly outperforms baselines, proving that adding structured semantic knowledge is a promising pathway for interpreting metaphorical language in risk detection.

1 Introduction

With over 700,000 annual deaths, suicide is a major public health crisis [World Health Organization, 2025]. Many individuals express distress on social media, creating a critical opportunity for early identification using computational models. However, automated detection is challenged by implicit and metaphorical language (e.g., "I want to sleep forever"), which standard models fail to interpret due to a lack of deep semantic understanding. Inspired by Cognitive Metaphor Theory (CMT) [Lakoff and Johnson, 1980], we address this gap by using Semantic Frames from FrameNet as a computationally tractable proxy to understand these metaphorical expressions. We hypothesize that frames can provide the deep semantic context that text-only models miss.

Therefore, this paper validates a deep learning framework for suicide risk classification that integrates semantic frame features. Our primary contributions are: 1. A Novel Framework combines FrameNet-derived features with textual embeddings in an end-to-end model. 2. Empirical Validation demonstrating superior performance against strong baselines on a public dataset. 3. A Practical Approach for leveraging an external knowledge base to better interpret non-literal expressions.

2 Related work

Early research on suicide risk detection relied on traditional supervised classifiers, employing logistic regression with user behavior features [Masuda et al., 2013], Support Vector Machines with psychological lexicons [Huang et al., 2014] and so on. Other work further validated a range of classifiers using diverse features, including statistical, linguistic, and topic features [Ji et al., 2018].

More recently, deep learning models have become the standard. Gkotsis et al. [2017] demonstrated an early application of CNN for 11 mental health analysis. CNN in Gaur et al. [2019]'s research outperformed in accurately distinguishing suicide risk levels compared with traditional models. Husseini Orabi et al. [2018] utilized a Bi-LSTM model with a context-aware attention mechanism generates the feature vector. Coppersmith et al. [2018] used similar method for long-term suicide risk scoring. Based on this, Sawhney et al. [2021] introduced ordinal regression layer that enables the model to learn the inherent sequential order of the risk levels. Advanced approaches now leverage Transformer-based models, either through ensemble with Negative Correlation Learning [Ragheb et al., 2023] or by integrating Large Language Models via voting mechanisms [Cao et al., 2024].

In NLP, metaphor processing is a distinct field, theoretically grounded in Conceptual Metaphor Theory (CMT) [Lakoff and Johnson, 1980], which conceptualizes abstract domains via concrete ones. In recent research, Stowe et al. [2021] have used conceptual mappings to produce metaphorical outputs, and Li et al. [2023] have enhanced detection performance by fine-tuning BERT with external knowledge from FrameNet to overcome shallow semantic limitations.

Despite extensive research in suicide risk identification and metaphor processing, few studies have applied principles from cognitive linguistics or knowledge bases to address implicit expressions. This paper is positioned at this intersection, exploring semantic frames as a lightweight knowledge injection method to bridge this semantic gap.

3 Methodology

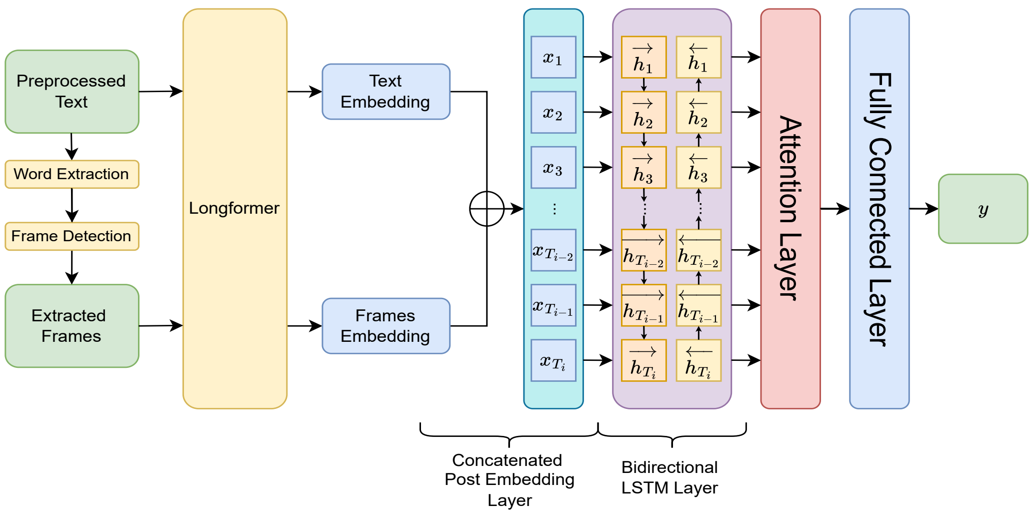

Figure 1: An overview of the proposed model architecture for suicide risk detection.

3.1 Problem Formulation

Given user $u_i \in U = \{u_1, u_2, \ldots, u_n\}$, our objective is to evaluate their suicide risk level $y \in Y$. Here, $Y = \{l_1, l_2, \ldots, l_k\}$ is a pre-defined set of k ordinal risk categories, representing increasing levels of severity. Each user u_i has a sequence of posts $P_i = \{p_i^1, p_i^2, \ldots, p_i^{T_i}\}$ over time, where p_i^t denotes the user's t^{th} post, with $p_i^{T_i}$ is the most recent post and T_i is the user's total number of posts.

3.2 Data

Dataset: This study utilizes dataset from Gaur et al. [2019]. Its gold-standard subset was annotated by practicing psychiatrists using the Columbia-Suicide Severity Rating Scale (C-SSRS), providing a 5-label ordinal classification scheme: Supportive (SU, 0), Indicator (IN, 1), Ideation (ID, 2), Behavior (BR, 3), and Attempt (AT, 4).

Data Preprocessing & Data Split: Data preprocessing included removing personally identifiable information, lowercasing, and eliminating punctuation and excess whitespace. Crucially, stop words conveying negation (e.g., 'not') were retained to preserve semantic meaning in critical expressions. The dataset was partitioned into training and testing sets with an 80:20 ratio.

Frame extraction & Post embedding: We use SpaCy to extract verbs and adjectives from the text to query the FrameNet knowledge base, concatenating the most frequent resulting frames into a single "frame-string". Both the original text and this frame-string are then independently encoded using a Longformer model. Their respective [CLS] token vectors serve as the final textual and semantic embeddings.

3.3 Model Architecture

Our model architecture consists of three key components: a Bi-LSTM layer, an attention mechanism, and a classification head, designed to capture temporal dependencies and post importance.

First, the Bi-LSTM layer processes the sequence of embedded posts $X = (x_1, x_2, \ldots, x_{T_i})$. By processing the sequence in both forward and backward directions, generating a contextualized hidden state h_t for each post, encoding information from both past and future posts:

$$\overrightarrow{h_t} = \text{LSTM}(x_t, \overrightarrow{h_{t-1}}) \qquad \overleftarrow{h_t} = \text{LSTM}(x_t, \overleftarrow{h_{t+1}}) \qquad h_t = [\overrightarrow{h_t}; \overleftarrow{h_t}]$$

Next, the attention mechanism computes a context vector c as a weighted sum of all hidden states $H = (h_1, h_2, \ldots, h_{T_i})$. This allows the model to assign higher weights (α_t) to key posts, with attention score e_t represents the un-normalized importance of each post:

$$e_t = \text{ReLU}(h_t W_a + b_a) \qquad \alpha_t = \frac{\exp(e_t)}{\sum_{j=1}^{T_i} \exp(e_j)} \qquad c = \sum_{t=1}^{T_i} \alpha_t h_t$$

Finally, this context vector c, representing the user's entire post history, is passed through a classification head (two fully-connected layers with Dropout) to produce the final risk prediction.

3.4 Training Details

We used the `Longformer-base-4096` model for embedding, trained with the AdamW optimizer. For the loss function, we employed an ordinal regression approach inspired by Sawhney et al. [2021], which applies a distance-weighted penalty (factor of 1.8) to predictions far from the true label. Key hyperparameters are listed in Table 1. To ensure stability, we report the median F1-Score over five runs for each model.

Table 1: Hyperparameter settings used for model training.

Hyperparameter	**Value**
Learning Rate	0.002
Dropout	0.3
LSTM Hidden Dimension	512
Number of LSTM Layers	1
Training Epochs	50
Batch Size	8
Early Stopping Patience	10

4 Empirical Analysis

4.1 Evaluation Metrics

To account for the severity of misclassifications, our evaluation follows metrics in Gaur et al. [2019]. Given a predicted risk level (r^p), an actual risk level (r^a) and the size of test data (N_T), we define a False Positive (FP) as the ratio of the number of time $r^p > r^a$, and a False Negative (FN) as the ratio of the number of time $r^p < r^a$. Additionally, we measure Ordinal Error (OE) as the ratio of samples where $|r^a - r^p|$ is greater than one. These components are then used to compute Precision, Recall, and the F-Score. The formal definitions of FP, FN and OE are as follows:

$$FP = \frac{\sum_{i=1}^{N_T} I(r_i^p > r_i^a)}{N_T}, \quad FN = \frac{\sum_{i=1}^{N_T} I(r_i^a > r_i^p)}{N_T}, \quad OE = \frac{\sum_{i=1}^{N_T} I(|r_i^a - r_i^p| > 1)}{N_T}$$

4.2 Quantitative Results

For evaluation, we consider two learning models (XGBoost [Ghosal and Jain, 2023] and Longformer [Beltagy et al., 2020]) as baselines. As shown in Table 2, our framework's key contribution is demonstrated by comparing our full model (Text+Frames) to its ablation variant (Text only). The integration of semantic frames led to an increase in Graded Recall from 0.49 to 0.53 and a sharp

Table 2: Experiment with 5-label Classification

Model	Input	Graded Precision	Graded Recall	F-Score	OE
XGBoost	TF-IDF vector	0.48	0.45	0.46	0.31
Longformer	Text+Frames	0.43	**0.53**	0.47	0.35
Bi-LSTM+Attention	Text+Frames	0.58	**0.53**	**0.55**	**0.15**
Bi-LSTM+Attention	Text	**0.62**	0.49	**0.55**	0.23

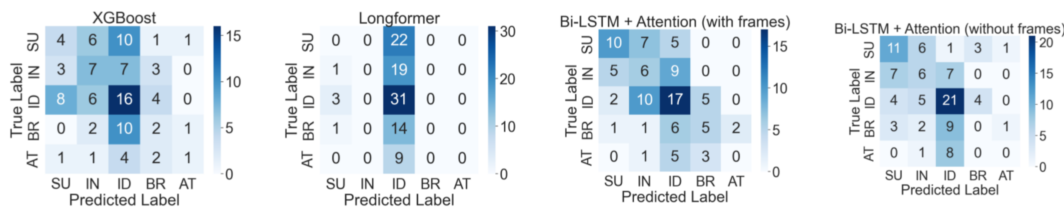

Figure 2: Confusion matrix of 4 models

reduction in OE from 0.23 to 0.15. This indicates our model is not only better at identifying at-risk users but also makes less severe misclassifications.
The confusion matrices in Figure 2 offer a qualitative explanation for this improvement. The baseline models exhibit predictable weaknesses. The XGBoost model, which relies on TF-IDF vectors, shows high confusion across all classes, likely because it represents the most common class with the most distinct vocabulary. The Longformer baseline, with its deep contextual understanding, though distinguish the most common ID perfectly, completely fails on the more critical category like BR and AT, showing struggles with the severe class imbalance if not make domain-specific fine-tuning. In contrast, the proposed model predict most of the AT examples as at risk labels. We attribute this success to the injection of semantic frames.

4.3 Qualitative Results

Table 3: Frames extraction from a case study

Text	True Label	Extracted Frames	Predicted Label
"I just want to die somewhere … She gets angry at me, destroys me, breaks me … Tired rarely she gives me some love … I'm Tired of this world."	Attempt	Emotion_directed Personal_relationship Desirability	Ideation Ideation **Behavior*** Ideation

Note: The predicted labels correspond to the following models, in order: XGBoost, Longformer, our proposed model (with frames), and the ablation model (without frames). The asterisk (*) denotes the best prediction.

The case study in Table 3 vividly illustrates this mechanism. For an "Attempt" user, baseline models misclassify the post as lower-risk "Ideation". However, our framework extracts high-level semantic frames (Emotion_directed, Personal_relationship, Desirability), allowing it to better grasp the situation's severity and predict a less severe "Behavior".

5 Discussion and Conclusion

In this work, we proposed a novel framework inspired by CMT and demonstrated that integrating semantic frame features with textual embeddings significantly improves the detection of suicidal ideation. A direction for future work is creating a benchmark dataset, which enables more advanced approaches like employing multi-task learning or combining to large language models.

Finally, it is essential that this framework should be regarded strictly as an assistive tool for human experts, not an autonomous diagnostic system, and should be applied with rigorous attention to potential model biases and data privacy concerns.

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Appendix: Code Availability

The full implementation of the model, along with training scripts and pre-processing tools, is publicly available at: `https://github.com/yw041202/DSAA2012_Project.git`.